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United States Patent	12413204
Kind Code	B2
Date of Patent	September 09, 2025
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Acoustic wave device, filter, and multiplexer

Abstract

An acoustic wave device includes a first piezoelectric layer that is a rotated Y-cut lithium tantalate substrate and has a first thickness, a second piezoelectric layer that is a rotated Y-cut lithium niobate substrate, is stacked on the first piezoelectric layer, has a second thickness that is less than the first thickness, and has a spontaneous polarization direction that is substantially opposite to a spontaneous polarization direction of the first piezoelectric layer, a first electrode provided on an opposite surface of the first piezoelectric layer from the second piezoelectric layer, and a second electrode that is provided on an opposite surface of the second piezoelectric layer from the first piezoelectric layer, at least a part of the first piezoelectric layer and at least a part of the second piezoelectric layer being interposed between the first electrode and the second electrode.

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Appl. No.: **18/170944**

Filed: **February 17, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230275566 A1	Aug. 31, 2023

Foreign Application Priority Data

JP	2022-027876	Feb. 25, 2022

Publication Classification

Int. Cl.: **H03H9/17** (20060101); **H03H9/02** (20060101); **H03H9/13** (20060101); **H03H9/56** (20060101); **H03H9/70** (20060101)

U.S. Cl.:

CPC **H03H9/173** (20130101); **H03H9/133** (20130101); **H03H9/568** (20130101); **H03H9/706** (20130101); H03H9/02031 (20130101)

Field of Classification Search

CPC: H03H (9/173); H03H (9/133); H03H (9/568); H03H (9/706); H03H (9/02031); H03H (9/02157); H03H (9/175); H03H (9/02015)

USPC: 333/187; 333/188

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S64-71207	12/1988	JP	N/A
H3-123214	12/1990	JP	N/A
H7-254836	12/1994	JP	N/A
H10-51262	12/1997	JP	N/A
2004-146861	12/2003	JP	N/A

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Background/Summary

FIELD

(1) A certain aspect of the present disclosure relates to an acoustic wave device, a filter, and a multiplexer.

BACKGROUND

(2) Bulk acoustic wave (BAW) resonators such as film bulk acoustic resonators (FBARs) and solid mounted resonators (SMRs) are used as filters and duplexers for high-frequency circuits of wireless terminals such as mobile phones. The BAW resonator is called a piezoelectric thin film resonator. The piezoelectric thin film resonator has a structure in which a pair of electrodes are provided with a piezoelectric layer interposed therebetween, and a resonance region where the pair of electrodes face each other across at least a part of the piezoelectric layer is a region where acoustic waves resonate. It is known that two piezoelectric layers having opposite spontaneous polarization

directions are stacked as a piezoelectric layer as disclosed in, for example, Japanese Patent Application Laid-Open Nos. S64-71207, 549-25883, H3-123214, H10-51262, and H7-254836 (Patent Documents 1 to 5, respectively). It is known to use a monocrystalline lithium niobate substrate and a monocrystalline lithium tantalate substrate as the two piezoelectric layers to be stacked (for example, Patent Document 5).

RELATED ART DOCUMENTS

Patent Documents

(3) Japanese Patent Application Laid-Open No. S64-71207 Japanese Patent Application Laid-Open No. S49-25883 Japanese Patent Application Laid-Open No. H3-123214 Japanese Patent Application Laid-Open No. H10-51262 Japanese Patent Application Laid-Open No. H7-254836

SUMMARY

(4) By stacking piezoelectric layers with spontaneous polarization directions opposite to each other, it is possible to excite the second-harmonic acoustic wave. When the second-harmonic acoustic wave is used as the main mode, the fundamental wave becomes an unnecessary wave. In the case that one of the stacked piezoelectric layers is a rotated Y-cut lithium niobate substrate and the other is a rotated Y-cut lithium tantalate substrate, it is required to inhibit excitation of the fundamental wave, which is an unnecessary wave.

(5) An object of the present disclosure is to reduce unnecessary waves.

(6) In one aspect of the present disclosure, there is provided an acoustic wave device including: a first piezoelectric layer that is a rotated Y-cut lithium tantalate substrate and has a first thickness; a second piezoelectric layer that is a rotated Y-cut lithium niobate substrate, is stacked on the first piezoelectric layer, has a second thickness that is less than the first thickness, and has a spontaneous polarization direction that is substantially opposite to a spontaneous polarization direction of the first piezoelectric layer; a first electrode provided on an opposite surface of the first piezoelectric layer from the second piezoelectric layer; and a second electrode that is provided on an opposite surface of the second piezoelectric layer from the first piezoelectric layer, at least a part of the first piezoelectric layer and at least a part of the second piezoelectric layer being interposed between the first electrode and the second electrode.

(7) In the above acoustic wave device, the first piezoelectric layer may be a 1580 or greater and 168° or less rotated Y-cut lithium tantalate substrate, the second piezoelectric layer may be a 1580 or greater and 168° or less rotated Y-cut lithium niobate substrate, and the second thickness may be equal to or greater than 0.5 times the first thickness.

(8) In the above acoustic wave device, the second thickness may be equal to or greater than 0.56 times the first thickness and is equal to or less than 0.92 times the first thickness.

(9) In the above acoustic wave device, the second thickness may be equal to or greater than 0.68 times the first thickness and is equal to or less than 0.78 times the first thickness.

(10) In the above acoustic wave device, an angle between an X-axis orientation of crystal orientations of a rotated Y-cut lithium tantalate substrate in the first piezoelectric layer and an X-axis orientation of crystal orientations of a rotated Y-cut lithium niobate substrate in the second piezoelectric layer may be 5° or less.

(11) In the above acoustic wave device, the first piezoelectric layer and the second piezoelectric layer may be directly bonded to each other.

(12) The above acoustic wave device may further include a support substrate, and the first piezoelectric layer may be provided between the support substrate and the second piezoelectric layer.

(13) The above acoustic wave device may further include a support substrate, and the second piezoelectric layer may be provided between the support substrate and the first piezoelectric layer.

(14) In another aspect of the present disclosure, there is provided a filter including the above acoustic wave device.

(15) In another aspect of the present disclosure, there is provided a multiplexer including the above filter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is a plan view of a piezoelectric thin film resonator in accordance with a first embodiment, and FIG. **1**B is a cross-sectional view taken along a line A-A in FIG. **1**A;
(2) FIG. **2**A and FIG. **2**B are views for describing the operations of the piezoelectric thin film resonators in a first comparative example and the first embodiment, respectively;
(3) FIG. **3** illustrates the crystal orientations of a piezoelectric layer in the first embodiment;
(4) FIG. **4** is a graph of |Y| versus frequency in a second comparative example;
(5) FIG. **5** is a graph of |Y| versus frequency in the first embodiment;
(6) FIG. **6**A is a graph of k.sup.2 of a fundamental wave versus Ta/Tb, and FIG. **6**B is a graph of Δk.sup.2 of the fundamental wave versus Ta/Tb;
(7) FIG. **7** is a graph of k.sup.2 of the fundamental wave versus Ta/Tb;
(8) FIG. **8** presents k.sup.2 and k.sup.2/k.sup.2(Ta/Tb=1) with respect to Ta/Tb;
(9) FIG. **9**A and FIG. **9**B are graphs of Ta/Tb versus θ;
(10) FIG. **10**A to FIG. **10**C are cross-sectional views of piezoelectric thin film resonators in accordance with first to third variations of the first embodiment, respectively;
(11) FIG. **11** is a cross-sectional view of a piezoelectric thin film resonator in accordance with a fourth variation of the first embodiment; and
(12) FIG. **12**A is a circuit diagram of a filter in accordance with a second embodiment, and FIG. **12**B is a circuit diagram of a duplexer in accordance with a first variation of the second embodiment.
DETAILED DESCRIPTION
(13) Hereinafter, embodiments of the present disclosure will be described with reference to the drawings.
First Embodiment
(14) A piezoelectric thin film resonator will be described as an example of the acoustic wave device.
(15) FIG. **1**A is a plan view of a piezoelectric thin film resonator in accordance with a first embodiment, and FIG. **1**B is a cross-sectional view taken along line A-A in FIG. **1**A. A stacking direction of piezoelectric layers **14***a* and **14***b* is defined as a Z direction, a direction in which a lower electrode **12** is extracted is defined as an X direction, and a direction orthogonal to the X direction and the Z direction is defined as a Y direction.
(16) As illustrated in FIG. **1**A and FIG. **1**B, a piezoelectric layer **14** is provided on a substrate **10**. The piezoelectric layer **14** includes the piezoelectric layers **14***a* and **14***b* that are stacked. The thicknesses of the piezoelectric layers **14***a* and **14***b* are Ta and Tb, respectively. The thickness of the piezoelectric layer **14** is T. The piezoelectric layer **14***a* is a rotated Y-cut lithium niobate (LiNbO.sub.3) substrate, and the piezoelectric layer **14***b* is a rotated Y-cut lithium tantalate (LiTaO.sub.3) substrate. A polarization direction **52***a* of the piezoelectric layer **14***a* is the downward direction. A polarization direction **52***b* of the piezoelectric layer **14***b* is the upward direction, and the piezoelectric layers **14***a* and **14***b* are directly bonded to each other using, for example, a surface-activation method. When the piezoelectric layers **14***a* and **14***b* are bonded to each other using the surface-activation method, an amorphous layer may be formed between the piezoelectric layers **14***a* and **14***b*. In this case, since the thickness of the amorphous layer is sufficiently smaller than those of the piezoelectric layers **14***a* and **14***b*, it can be considered that the piezoelectric layers **14***a* and **14***b* are directly bonded to each other.

(17) An upper electrode **16** and the lower electrode **12** are provided on and below the piezoelectric layer **14**, respectively. A region where the lower electrode **12** and the upper electrode **16** overlap in a plan view with at least a part of the piezoelectric layer **14***a* and at least a part of the piezoelectric layer **14***b* interposed therebetween in the stacking direction is a resonance region **50**. An air gap **34** is provided between the substrate **10** and the lower electrode **12**. The acoustic wave is reflected by the interface between the lower electrode **12** and the air gap **34**. In a plan view, the air gap **34** overlaps with the resonance region **50**, and the air gap **34** has the same size as the resonance region **50** or is larger than the resonance region **50**.

(18) The substrate **10** is, for example, a silicon substrate, a sapphire substrate, an alumina substrate, a spinel substrate, a quartz substrate, a crystal substrate, a glass substrate, a ceramic substrate, a GaAs substrate, or the like. Each of the lower electrode **12** and the upper electrode **16** is, for example, a single-layer film of ruthenium (Ru), chromium (Cr), aluminum (Al), titanium (Ti), copper (Cu), molybdenum (Mo), tungsten (W), tantalum (Ta), platinum (Pt), rhodium (Rh), iridium (Ir), or the like, or a multilayer film in which these films are stacked.

(19) FIG. **2**A and FIG. **2**B are views for describing the operations of the piezoelectric thin film resonators in a first comparative example and the first embodiment, respectively. In FIG. **2**A and FIG. **2**B, the lower electrode **12**, the piezoelectric layer **14**, and the upper electrode **16** are illustrated. The graph on the right side illustrates the displacement of the acoustic wave with respect to the position in the Z direction. In FIG. **2**B, the thicknesses Ta and Tb are illustrated as being equal to each other for the sake of simplicity.

(20) As illustrated in FIG. **2**A, in the first comparative example, the piezoelectric layer **14** is one piezoelectric layer in which a polarization direction **52** is one direction. In this case, as illustrated in the graph on the right side, in the fundamental wave, when the displacement of the upper surface of the piezoelectric layer **14** is 0, the displacement of the lower surface of the piezoelectric layer **14** becomes E. Thus, in the fundamental wave, the potential of the lower electrode **12** and the potential of the upper electrode **16** are different from each other, and when an AC signal having a predetermined frequency is applied between the lower electrode **12** and the upper electrode **16**, the fundamental wave is excited in the piezoelectric layer **14**. On the other hand, in the second harmonic, the potential of the lower electrode **12** and the potential of the upper electrode **16** are substantially the same. Therefore, the second harmonic is hardly excited in the piezoelectric layer **14**. The thickness T of the piezoelectric layer **14** is approximately ½ of the wavelength of the fundamental wave. To manufacture a piezoelectric thin film resonator having a high resonant frequency, the thickness T of the piezoelectric layer **14** is reduced. However, since the air gap **34** exists between the piezoelectric layer **14** and the support substrate, when the piezoelectric layer **14** is made thin, the piezoelectric layer **14** is likely to break. In addition, because of manufacturing variations in the thickness T of the piezoelectric layer **14**, characteristics such as the resonant frequency are likely to vary.

(21) As illustrated in FIG. **2**B, in the first embodiment, the polarization direction **52***a* of the piezoelectric layer **14***a* is the downward direction, and the polarization direction **52***b* of the piezoelectric layer **14***b* is the upward direction. As illustrated in the graph on the right side, in the second harmonic, when the displacement of the upper surface of the piezoelectric layer **14** is 0, the displacement of the lower surface of the piezoelectric layer **14** is 0. The displacement near the center of the piezoelectric layer **14** in the Z direction is E. Since the polarization direction **52***a* of the piezoelectric layer **14***a* and the polarization direction **52***b* of the piezoelectric layer **14***b* are opposite to each other, the potentials of the lower electrode **12** and the upper electrode **16** are different from each other in the second harmonic. Therefore, when an AC current having a predetermined frequency is applied between the lower electrode **12** and the upper electrode **16**, the second harmonic is excited. On the other hand, in the fundamental wave, the potential of the lower electrode **12** and the potential of the upper electrode **16** are substantially the same. Therefore, the fundamental wave is hardly excited in the piezoelectric layer **14**. The thickness T is approximately

the wavelength of the second harmonic. For this reason, when the piezoelectric thin film resonator having the same resonant frequency as the piezoelectric film of the first comparative example is manufactured, the thickness T of the piezoelectric layer **14** in the first embodiment can be about twice that in the first comparative example. As a result, breakage of the piezoelectric layer **14** can be inhibited. In addition, variations in characteristics such as a resonant frequency can be reduced.
(22) When lithium tantalate substrates having a rotation cut angle of 163° or lithium niobate substrates having a rotation cut angle of 163° is used for the piezoelectric layers **14***a* and **14***b*, thickness-shear vibration is excited in the piezoelectric layer **14**. The vibration direction of the thickness shear vibration is the Y direction. In addition, thickness extension vibration is hardly excited in the piezoelectric layer **14**. Therefore, when the thickness shear vibration is the main mode, the rotation cut angle is preferably 163°.
(23) When the piezoelectric layers **14***a* and **14***b* are made of the same material, the piezoelectric characteristics are determined by the material. Therefore, by adopting a lithium tantalate substrate as the piezoelectric layer **14***a* and a lithium niobate substrate as the piezoelectric layer **14***b*, the piezoelectric characteristics such as the electromechanical coupling coefficient k.sup.2 can be adjusted to be between the piezoelectric characteristics obtained when the piezoelectric layers **14***a* and **14***b* are lithium tantalate substrates and the piezoelectric characteristics obtained when the piezoelectric layers **14***a* and **14***b* are lithium niobate substrates.
(24) FIG. **3** illustrates the crystal orientations of the piezoelectric layer in the first embodiment. The X-axis orientation, the Y-axis orientation, and the Z-axis orientation are axial orientations of the crystal orientations, and the X direction, the Y direction, and the Z direction are directions presented in FIG. **1**A and FIG. **1**B. A 163° rotated Y-cut lithium niobate substrate is used as the piezoelectric layer **14***a*, and a 163° rotated Y-cut lithium tantalate substrate is used as the piezoelectric layer **14***b*.
(25) As illustrated in FIG. **3**, the +X-axis orientation of the crystal orientations of the piezoelectric layer **14***a* and the +X-axis orientation of the crystal orientations of the piezoelectric layer **14***b* are substantially the same orientation, which is the X direction. In the piezoelectric layer **14***a*, the +Y-axis orientation and the +Z-axis orientation are adjusted to be the −Z direction and the +Y direction, respectively, as indicated by broken line arrows. The Y-axis and the Z-axis are rotated around the X-axis by θ=163° from the +Z-axis orientation toward the +Y-axis orientation. The lithium tantalate substrate and the lithium niobate substrate have crystal structures similar to trigonal ilmenite, and the +Z-axis orientation thereof corresponds to the c-axis orientation of the crystal orientations. The +Z-axis orientation is the spontaneous polarization direction. In this case, the upper surface and the lower surface of the piezoelectric layer **14***a* are referred to as a negative (−) surface and a positive (+) surface, respectively, and the polarization direction **52***a* is illustrated downward.
(26) In the piezoelectric layer **14***b*, the +Y-axis orientation and the +Z-axis orientation are adjusted to be the +Z direction and the −Y direction, respectively, as indicated by dashed arrows. The Y-axis and the Z-axis are rotated around the X-axis by θ=163° from the +Z-axis orientation toward the +Y-axis orientation. The +Z-axis orientation in this case corresponds to the c-axis orientation of the crystal orientations. The +Z-axis orientation is the spontaneous polarization direction. In this case, the upper surface and the lower surface of the piezoelectric layer **14***b* are referred to as a positive (+) surface and a negative (−) surface, respectively, and the polarization direction **52***b* is illustrated upward. The spontaneous polarization directions of the piezoelectric layers **14***a* and **14***b* can be confirmed by identifying the Z-axis orientation using an X-ray diffraction method.
(27) Spontaneous polarization is a state of polarization in a state where no electric field is applied. In lithium tantalate, Li and Ta are cations, and O is an anion. If atoms of Li, Ta, and O are positioned so as to cancel electric charges, spontaneous polarization does not occur. Since these atoms are displaced from the positions where the electric charges are canceled out, spontaneous polarization occurs. Spontaneous polarization also occurs in lithium niobate.

(28) By adjusting the +X-axis orientation of the piezoelectric layer **14***a* and the +X-axis orientation of the piezoelectric layer **14***b* to be substantially the same direction, the spontaneous polarization direction (+Z-axis orientation) of the piezoelectric layer **14***a* and the spontaneous polarization direction (+Z-axis orientation) of the piezoelectric layer **14***b* become substantially opposite directions. For example, when the +X-axis orientation of the piezoelectric layer **14***a* and the X-axis orientation of the piezoelectric layer **14***b* are different from each other, the spontaneous polarization direction (+Z-axis orientation) of the piezoelectric layer **14***a* and the spontaneous polarization direction (+Z-axis orientation) of the piezoelectric layer **14***b* are not opposite to each other. When the spontaneous polarization direction (+Z-axis orientation) of the piezoelectric layer **14***a* and the spontaneous polarization direction (+Z-axis orientation) of the piezoelectric layer **14***b* are substantially opposite to each other, this means it is acceptable that an angle between the +X-axis orientation of the piezoelectric layer **14***a* and the +X-axis orientation of the piezoelectric layer **14***b* is within a range of 5° or less.
(29) Simulation
(30) The frequency characteristics of the piezoelectric thin film resonator were simulated using a finite element method. The simulation conditions are as follows. Thickness T of the piezoelectric layer **14**: 300 nm Lower electrode **12**: Aluminum film with a thickness of 30 nm Upper electrode **16**: Aluminum film with a thickness of 30 nm
(31) The ratio of the thicknesses Ta of the piezoelectric layers **14***a* to the thicknesses Tb of the piezoelectric layers **14***b* was defined as Ta/Tb. When Ta/Tb=1, the thicknesses Ta and Tb of the piezoelectric layers **14***a* and **14***b* are both 150 nm.
(32) As a second comparative example, frequency characteristics in a case in which 163° rotated Y-cut lithium niobate substrates were used for both the piezoelectric layers **14***a* and **14***b* were simulated. The piezoelectric layer **14***a* and the piezoelectric layer **14***b* are bonded to each other so that polarization directions thereof are opposite to each other. FIG. **4** is a graph of |Y| versus frequency in the second comparative example. Here, |Y| is the absolute value of the admittance. As presented in FIG. **4**, when Ta/Tb=1, a resonant frequency fr**2** and an antiresonant frequency fa**2** are observed as the response of the acoustic wave due to the second harmonic. No response of the fundamental wave is observed. When Ta/Tb=0.9048, the resonant frequency fr**2** and the antiresonant frequency fa**2** are observed as the response of the acoustic wave due to the second harmonic, and a resonant frequency fr**1** and an antiresonant frequency fa**1** are observed as the response of the acoustic wave due to the fundamental wave.
(33) When the piezoelectric layers **14***a* and **14***b* are made of the same material as in the second comparative example, the fundamental wave, which is an unnecessary wave, is hardly excited by setting Ta/Tb=1. On the other hand, when Ta/Tb is set to a value different from 1, the fundamental wave, which is an unnecessary wave, is excited.
(34) As the first embodiment, frequency characteristics in a case in which a 163° rotated Y-cut lithium niobate substrate was used for the piezoelectric layer **14***a* and a 163° rotated Y-cut lithium tantalate substrate was used for the piezoelectric layer **14***b* were simulated. The piezoelectric layer **14***a* and the piezoelectric layer **14***b* are bonded to each other so that polarization directions thereof are opposite to each other. FIG. **5** is a graph of |Y| versus frequency in the first embodiment. Here, |Y| is the absolute value of the admittance. As presented in FIG. **5**, when Ta/Tb=1, the resonant frequency fr**2** and the antiresonant frequency fa**2** are observed as the response of the acoustic wave due to the second harmonic, and the resonant frequency fr**1** and the antiresonant frequency fa**1** are observed as the response of the acoustic wave due to the fundamental wave. When Ta/Tb=0.7178, the resonant frequency fr**2** and the antiresonant frequency fa**2** are observed as the response of the acoustic wave due to the second harmonic, and the response of the fundamental wave is not observed.
(35) As described above, when a rotated Y-cut lithium niobate substrate is used for the piezoelectric layer **14***a* and a rotated Y-cut lithium tantalate substrate is used for the piezoelectric layer **14***b*, the

fundamental wave can be suppressed by adjusting the thicknesses Ta to be smaller than the thicknesses Tb. Therefore, the electromechanical coupling coefficient k.sup.2 of the fundamental wave was simulated for different Ta/Tb.

(36) FIG. **6**A is a graph of k.sup.2 of the fundamental wave versus Ta/Tb, and FIG. **6**B is a graph of Δk.sup.2 of the fundamental wave versus Ta/Tb. In FIG. **6**A, black circles indicate simulated points, and a curve is a line connecting the black circles. In FIG. **6**B, Δk.sup.2 is the rate of change of k.sup.2 with respect to Ta/Tb, and corresponds to the slope between the adjacent black circles in FIG. **6**A.

(37) As illustrated in FIG. **6**A and FIG. **6**B, the electromechanical coupling coefficient k.sup.2 of the fundamental wave is about 0.06 in a range of Ta/Tb from 0.70 to 0.75. In a range of Ta/Tb of 0.7 or less, k.sup.2 increases as Ta/Tb decreases, and in a range of Ta/Tb of 0.75 or greater, k.sup.2 increases as Ta/Tb increases.

(38) By setting θ to 158°, 163° and 168° and varying Ta/Tb, k.sup.2 of the fundamental wave was simulated. FIG. **7** is a graph of k.sup.2 of the fundamental wave versus Ta/Tb. As presented in FIG. **7**, k.sup.2 of the fundamental wave is smallest in a range of Ta/Tb of 0.70 to 0.75 for any of θ of 158°, 163°, and 168°. Ranges R**1** to R**3** are ranges in which k.sup.2 of the fundamental wave is smaller than k.sup.2(Ta/Tb=1), which is k.sup.2 when Ta/Tb=1, when 0 is 158°, 163°, and 168°, respectively. When Ta/Tb is within the ranges R**1** to R**3**, k.sup.2 of the fundamental wave is equal to or less than k.sup.2(Ta/Tb=1), which is k.sup.2 when Ta/Tb=1.

(39) FIG. **8** presents k.sup.2 and k.sup.2/k.sup.2(Ta/Tb=1) with respect to Ta/Tb. For 0 of 158°, 163° and 168°, Ta/Tb at which k.sup.2/k.sup.2(Ta/Tb=1) is 1.0, 0.8, 0.6, 0.5, 0.4 and 0.2 and Ta/Tb at which k.sup.2 is 0.1% are presented.

(40) FIG. **9**A and FIG. **9**B are graphs of Ta/Tb versus θ. FIG. **9**A presents a range of Ta/Tb in which k.sup.2/k.sup.2(Ta/Tb=1) is smaller than 0.5, and FIG. **9**B presents a range in which k.sup.2 of the fundamental wave is 0.1% or less. In FIG. **9**A and FIG. **9**B, black circles indicate simulation points, straight lines indicate approximate straight lines of the black circles, and hatches indicate ranges between the straight lines.

(41) As presented in FIG. **9**A and FIG. **9**B, the range of Ta/Tb in which k.sup.2/k.sup.2(Ta/Tb=1) is smaller than 0.5, and the range of Ta/Tb in which k.sup.2 of the fundamental wave is 0.1% or less do not depend much on θ.

(42) In the first embodiment, as illustrated in FIG. **1**A and FIG. **1**B, the piezoelectric layer **14***b* (a first piezoelectric layer) and the piezoelectric layer **14***a* (a second piezoelectric layer) are stacked. The piezoelectric layer **14***b* is a rotated Y-cut lithium tantalate substrate and has a thickness Tb (a first thickness). The piezoelectric layer **14***a* is a rotated Y-cut lithium niobate substrate, has a thickness Ta (a second thickness), and has a spontaneous polarization direction substantially opposite to the spontaneous polarization direction of the piezoelectric layer **14***b*. The lower electrode **12** (a first electrode) is provided on the opposite surface of the piezoelectric layer **14***b* from the piezoelectric layer **14***a*. The upper electrode **16** (a second electrode) is provided on the opposite surface of the piezoelectric layer **14***a* from the piezoelectric layer **14***b*, and sandwich at least a part of the piezoelectric layer **14***b* and at least a part of the piezoelectric layer **14***a* between the upper electrode **16** and the lower electrodes **12**. In such a structure, the thickness Ta is adjusted to be smaller than the thickness Tb. This configuration can reduce k.sup.2 of the fundamental wave, thereby reducing unnecessary waves. Note that the term “the spontaneous polarization directions of the piezoelectric layers **14***a* and **14***b* are substantially opposite to each other” means the spontaneous polarization directions are opposite to each other to such an extent that the fundamental wave can be reduced. The angle between the spontaneous polarization directions of the piezoelectric layers **14***a* and **14***b* is preferably 1750 or greater and 1850 or less, more preferably 1770 or greater and 1830 or less, and further preferably 1780 or greater and 1820 or less.

(43) As can be seen from FIG. **8** and FIG. **9**B, the magnitude of the fundamental wave does not greatly depend on θ. Therefore, it is considered that the same results as in FIG. **8** to FIG. **9**B can be

obtained as long as the piezoelectric layer **14***b* is a 1580 or greater and 1680 or less rotated Y-cut lithium tantalate substrate and the piezoelectric layer **14***a* is a 1580 or greater and 1680 or less rotated Y-cut lithium niobate substrate. From FIG. **8**, when the thickness Ta is equal to or greater than 0.5 times the thickness Tb, k.sup.2/k.sup.2(Ta/Tb=1) can be adjusted to be smaller than 1. In order to excite the acoustic wave of thickness-shear vibration and not to excite the acoustic wave of thickness extension vibration, θ of the piezoelectric layers **14***b* and **14***a* is preferably 1600 or greater and 1660 or less, and more preferably 161° or greater and 165° or less.

(44) In FIG. **8**, when Ta/Tb is 0.52 or greater and 0.97 or less, k.sup.2/k.sup.2(Ta/Tb=1) can be adjusted to be 0.8 or less. When Ta/Tb is 0.55 or greater and 0.94 or less, k.sup.2/k.sup.2(Ta/Tb=1) can be adjusted to be 0.6 or less. When Ta/Tb is 0.56 or greater and 0.92 or less, k.sup.2/k.sup.2(Ta/Tb=1) can be adjusted to be 0.5 or less. When Ta/Tb is 0.58 or greater and 0.90 or less, k.sup.2/k.sup.2(Ta/Tb=1) can be adjusted to be 0.4 or less. When Ta/Tb is 0.62 or greater and 0.85 or less, k.sup.2/k.sup.2(Ta/Tb=1) can be adjusted to be 0.2 or less. When Ta/Tb is 0.68 or greater and 0.78 or less, k.sup.2 can be adjusted to be 0.1% or less.

(45) To make the spontaneous polarization direction of the piezoelectric layer **14***a* and the spontaneous polarization direction of the piezoelectric layer **14***b* substantially opposite to each other, the angle between the X-axis orientation of the crystal orientations of the lithium tantalate substrate in the piezoelectric layer **14***a* and the X-axis orientation of the crystal orientations of the lithium niobate substrate in the piezoelectric layer **14***b* is preferably 5° or less, more preferably 2° or less.

(46) The piezoelectric layers **14***a* and **14***b* are directly bonded to each other, and an interface between the piezoelectric layers **14***a* and **14***b* is flat (for example, the arithmetic surface roughness Ra is 10 nm or less). Thereby, the second harmonic can be excited in the piezoelectric layer **14**.

First Variation of the First Embodiment

(47) FIG. **10**A to FIG. **10**C are cross-sectional views of piezoelectric thin film resonators in accordance with first to third variations of the first embodiment, respectively. As illustrated in FIG. **10**A, in the first variation of the first embodiment, the polarization direction **52***a* of the piezoelectric layer **14***a* is the upward direction, and the polarization direction **52***b* of the piezoelectric layer **14***b* is the downward direction. Other configurations are the same as those of the first embodiment, and a description thereof is omitted.

Second Variation of the First Embodiment

(48) As illustrated in FIG. **10**B, in the second variation of the first embodiment, the upper surface of the piezoelectric layer **14***a* and the lower surface of the piezoelectric layer **14***b* are bonded to each other, the lower electrode **12** is provided on the lower surface of the piezoelectric layer **14***a*, and the upper electrode **16** is provided on the upper surface of the piezoelectric layer **14***b*. The polarization direction **52***a* of the piezoelectric layer **14***a* is the upward direction, and the polarization direction **52***b* of the piezoelectric layer **14***b* is the downward direction. Other configurations are the same as those of the first embodiment, and a description thereof is omitted.

Third Variation of the First Embodiment

(49) As illustrated in FIG. **10**C, in the third variation of the first embodiment, the polarization direction **52***a* of the piezoelectric layer **14***a* is the downward direction, and the polarization direction **52***b* of the piezoelectric layer **14***b* is the upward direction. Other configurations are the same as those of the second variation of the first embodiment, and the description thereof will be omitted.

(50) As in the first embodiment and the first variation thereof, the piezoelectric layer **14***b* may be provided between the substrate **10** (a support substrate) and the piezoelectric layer **14***a*. As in the second and third variations of the first embodiment, the piezoelectric layer **14***a* may be provided between the substrate **10** and the piezoelectric layer **14***b*. Ta/Tb with which the fundamental wave is reduced is determined by the amount of electric charge generated by spontaneous polarization when the fundamental wave is excited. Therefore, the preferable range of Ta/Tb obtained by the

above simulation can also be applied to the first to third variations of the first embodiment. The preferable range of Ta/Tb can be applied without depending on the frequencies such as the resonant frequency of the piezoelectric thin film resonator, the materials of the lower electrode **12** and the upper electrode **16**, and the piezoelectric layers **14***a* and **14***b* and the lower electrode **12** and the upper electrode **16**.

Fourth Variation of the First Embodiment

(51) FIG. **11** is a cross-sectional view of a piezoelectric thin film resonator in accordance with a fourth variation of the first embodiment. As illustrated in FIG. **11**, in the fourth variation of the first embodiment, an acoustic mirror **30** is provided between the substrate **10** and the lower electrode **12** instead of the air gap **34**. In the acoustic mirror **30**, a film **31** having low acoustic impedance and a film **32** having high acoustic impedance are alternately provided. The thickness of each of the films **31** and **32** is, for example, approximately λ/4 (λ is the wavelength of the acoustic wave). Thereby, the acoustic mirror **30** reflects the acoustic wave. The number of layers of the film **31** and the film **32** can be freely selected. The acoustic mirror **30** overlaps the resonance region **50**, and has the same size as the resonance region **50** or larger than the resonance region **50**. The film **31** of the acoustic mirror **30** is made of, for example, silicon oxide, silicon nitride, or the like. The film **32** is made of, for example, tungsten, tantalum, molybdenum, ruthenium, or the like.

(52) As in the first embodiment and the first to third variations thereof, the piezoelectric thin film resonator may be an FBAR in which the air gap **34** reflects an acoustic wave. As in the fourth variation of the first embodiment, the piezoelectric thin film resonator may be an SMR in which the acoustic mirror **30** reflects an acoustic wave. Although an example in which the planar shape of the resonance region **50** is rectangular has been described, the resonance region **50** may have an elliptical shape or a polygonal shape such as a pentagonal shape.

Second Embodiment

(53) A second embodiment is an exemplary filter and an exemplary duplexer using the piezoelectric thin film resonator of the first embodiment. FIG. **12**A is a circuit diagram of a filter in accordance with the second embodiment. As illustrated in FIG. **12**A, one or more series resonators S**1** to S**4** are connected in series between an input terminal Tin and an output terminal Tout. One or more parallel resonators P**1** to P**4** are connected in parallel between the input terminal Tin and the output terminal Tout. The piezoelectric thin film resonator of the first embodiment can be used for at least one of the following resonators: the one or more series resonators S**1** to S**4** and the one or more parallel resonators P**1** to P**4**. The number of resonators of the ladder-type filter can be freely selected.

(54) FIG. **12**B is a circuit diagram of a duplexer in accordance with a first variation of the second embodiment. As illustrated in FIG. **12**B, a transmit filter **40** is connected between a common terminal Ant and a transmit terminal Tx. A receive filter **42** is connected between the common terminal Ant and a receive terminal Rx. The transmit filter **40** transmits signals in the transmit band to the common terminal Ant as transmission signals among signals input from the transmit terminal Tx, and suppresses signals with other frequencies. The receive filter **42** transmits signals in the receive band to the receive terminal Rx as reception signals among signals input from the common terminal Ant, and suppresses signals with other frequencies. At least one of the transmit filter **40** or the receive filter **42** may be the filter of the second embodiment.

(55) Although the duplexer has been described as an example of the multiplexer, a triplexer or a quadplexer may be used.

(56) Although the embodiment of the present invention has been described in detail above, the present invention is not limited to the specific embodiment, and various modifications and changes can be made within the scope of the gist of the present invention described in the claims.

Claims

1. An acoustic wave device, comprising: a first piezoelectric layer that is a rotated Y-cut lithium tantalate substrate and has a first thickness; a second piezoelectric layer that is a rotated Y-cut lithium niobate substrate, is stacked on the first piezoelectric layer, has a second thickness that is less than the first thickness, and has a spontaneous polarization direction that is substantially opposite to a spontaneous polarization direction of the first piezoelectric layer; a first electrode provided on an opposite surface of the first piezoelectric layer from the second piezoelectric layer; and a second electrode that is provided on an opposite surface of the second piezoelectric layer from the first piezoelectric layer, at least a part of the first piezoelectric layer and at least a part of the second piezoelectric layer being interposed between the first electrode and the second electrode.
2. The acoustic wave device according to claim 1, wherein the first piezoelectric layer is a 1580 or greater and 1680 or less rotated Y-cut lithium tantalate substrate, wherein the second piezoelectric layer is a 1580 or greater and 1680 or less rotated Y-cut lithium niobate substrate, and wherein the second thickness is equal to or greater than 0.5 times the first thickness.
3. The acoustic wave device according to claim 2, wherein the second thickness is equal to or greater than 0.56 times the first thickness and is equal to or less than 0.92 times the first thickness.
4. The acoustic wave device according to claim 2, wherein the second thickness is equal to or greater than 0.68 times the first thickness and is equal to or less than 0.78 times the first thickness.
5. The acoustic wave device according to claim 2, wherein an angle between an X-axis orientation of crystal orientations of the rotated Y-cut lithium tantalate substrate in the first piezoelectric layer and an X-axis orientation of crystal orientations of the rotated Y-cut lithium niobate substrate in the second piezoelectric layer is 5° or less.
6. The acoustic wave device according to claim 1, wherein the first piezoelectric layer and the second piezoelectric layer are directly bonded to each other.
7. The acoustic wave device according to claim 1, further comprising: a support substrate, wherein the first piezoelectric layer is provided between the support substrate and the second piezoelectric layer.
8. The acoustic wave device according to claim 1, further comprising: a support substrate, wherein the second piezoelectric layer is provided between the support substrate and the first piezoelectric layer.
9. A filter comprising: the acoustic wave device according to claim 1.
10. A multiplexer comprising: the filter according to claim 9.